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Patched by Phone Call, Not by Score

Interviewing council road crews on how they actually sequence pothole repairs

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Background

Council road-maintenance apps rank reported potholes by a predicted-deterioration score and forecast next quarter's resurfacing schedule from it. Little is known about whether the crews holding the patch truck still work that ranking once they have spent a season reading the road themselves.

Aims

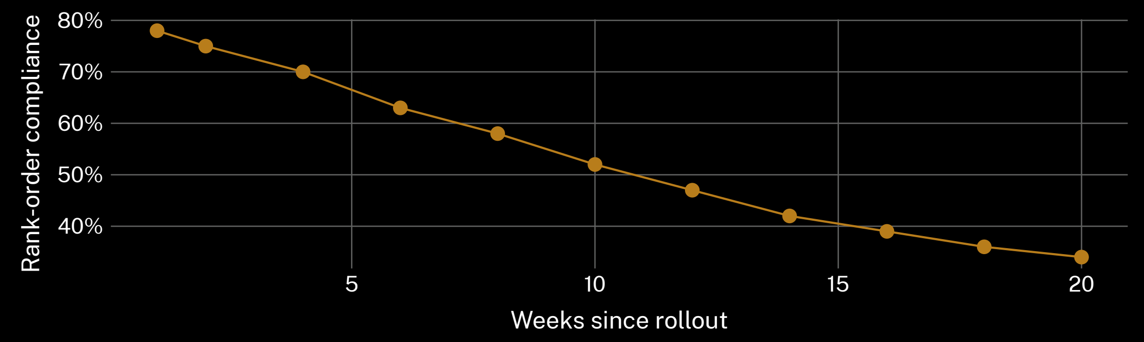
- Track how closely early-season patch order follows the app's predicted-deterioration rank
- Catalogue the cues crews substitute once the rank stops holding
- Test whether an app update restores compliance or the erosion continues regardless

Methods

- 34 crew members · 9 councils · semi-structured interviews + ride-alongs
- one wet season; 341 dispatch-log jobs coded against logged rank at time of patch
- patch-order rationale grounded-coded independently by two researchers; disagreements adjudicated blind to the app's rank

Results

Compliance with the app's rank order was highest in week 1 and fell steadily thereafter, tracking not the score but how long a crew had been on the road.



Share of weekly patch jobs actioned in the app's exact predicted-deterioration rank order fell from 78% in week 1 to 34% by week 20 across nine councils (n = 341 logged jobs).

Discussion & conclusions

Crews substitute cues the model never sees — complaint-call heat, school-run visibility, a subsidence "look" learned on the job — once the score stops matching what the road tells them. The rank's authority erodes almost exactly as fast as a crew learns which potholes actually get complained about. Next: whether feeding crew override reasons back into the model closes the gap, or just teaches it to mimic the queue crews already carry in their heads.

References

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Interview protocol approved under General Human Ethics 2025/162; crew rosters de-identified to council level only.

"The score says next. The phone says now."

Rank-order compliance fell from 78% to 34% across one wet season.